Hydrochemistry Report

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Experiment	Chemical De-acidification using calcium carbonate and semi-burnt dolomite
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Aim of the Experiment

To evaluate the efficiency of four different commercial filter materials (HYDROLIT-Ca, Hydro-Carbonate, AKDOLIT-GRAN and MAGNO-DOL) for the de-acidification of corrosive water. By passing the model water through calcium reactors we aim to establish calco-carbonic equilibrium. The study will measure how these reactors affect the pH, conductivity (K_{25C}), acid neutralizing capacity ($ANC_{4.3}$ and $ANC_{8.2}$) and the concentrations of Calcium and Magnesium ions.

AKDOLIT-GRAN and MAGNO-DOL is hypothesized to perform better. They are semi-calcined (heat treated) which converts Magnesium part of the rock into its Oxide which reacts vigorously and instantly with water and acid compared to other forms of carbonate materials based on $CaCO_3$. Compared to MAGNO-DOL, AKDOLIT-GRAN has a regular granular (spherical) shape which might help it react evenly due to its regular shape.

Theoretical Background

Definitions

- **Calco-carbonic equilibrium:** A state where water does not have excess calcium carbonate (corrosive) nor precipitate it (scaling). It is determined by the balance between dissolved Ca^{2+} ions and the carbonic acid (CO_2, HCO_3^-, CO_3^-).
- **Total Hardness:** The sum of alkaline earth metal concentrations, primarily calcium (Ca^{2+}) and (Mg^{2+}), dissolved in water.
- **Carbonate Hardness (Temporary Hardness):** The portion of hardness equivalent to the concentration of bicarbonate (HCO_3^-) and carbonate (CO_3^{2-}). It can be removed by boiling.
- **Permanent Hardness (Non-carbonate Hardness):** The portion of hardness associated with anions other than carbonates (e.g. sulfates, chlorides) which cannot be removed by boiling.
- **Acid Neutralizing Capacity (ANC):** The amount of strong acid required to titrate a water sample to a specific pH (usually 4.3). $ANC_{4.3}$ corresponds to the concentration of bicarbonate ions.
- **Base Neutralizing Capacity (BNC):** The amount of strong base required to titrate a sample to pH 8.2. $BNC_{8.2}$ corresponds to the concentration of free CO_2.

Chemical Equations for De-acidification: The de-acidification process involves the reaction of aggressive CO_2 with the filter materials.

- **Reaction with Calcium Carbonate ($CaCO_3$):** Occurs in all reactors (1, 2, 3, and 4) and also in the raw water sample.

$$CaCO_{3(s)} + CO_2 + H_2O \rightleftharpoons Ca^{2+} + 2\,HCO_3^-$$

- **Reaction with Magnesium Carbonate ($MgCO_3$):** Occurs primarily in Reactor 1 (HYDROLIT-Ca).

$$MgCO_{3(s)} + CO_2 + H_2O \rightleftharpoons Mg^{2+} + 2\,HCO_3^-$$

- **Reaction with Magnesium Oxide (MgO):** Occurs in Reactors 3 (AKDOLIT) and 4 (MAGNO-DOL).

$$MgO_{(s)} + 2CO_2 + H_2O \rightleftharpoons Mg^{2+} + 2\,HCO_3^-$$

Experiment

The experiment was conducted according to the procedure described in the script "MHSE 08 – Hydrochemistry, Experiment: Chemical De-acidification" **[1]**. No Significant deviations from the script were observed during the experiment

Results and Discussion

Evaluation of Personal Experimental Part (Raw Water)

My group was responsible for the analysis of the "**Raw Water**". The following raw titration data (volumes) were determined during the experiment and used to calculate the concentrations.

1) **Acid Neutralizing Capacity (ANC4.3):** The titration of 100 mL of raw water with 0.1M HCl consumed 2.3 mL of acid.

$$ANC_{4.3} = \frac{V_T * c_T}{V_S} * 1000 = \frac{2.3\,mL * 0.1\,mol/L}{100\,mL} * 1000 = 2.3\,mmol/L$$

This represents the concentration of bicarbonate (HCO_3^-).

2) **Base Neutralizing Capacity ($BNC_{8.2}$):** The titration of 100 mL of raw water with 0.1 M NaOH consumed 3.6 mL of base.

$$BNC_{8.2} = \frac{V_T * c_T}{V_S} * 1000 = \frac{3.6\,mL * 0.1\,mol/L}{100\,mL} * 1000 = 3.6 mmol/L$$

This represents the concentration of free carbon dioxide (CO_2)

3) **Calcium Concentration (c (Ca^{2+})):** The titration of 50 mL of raw water with 0.01 M EDTA (using Calcon-carbonic acid indicator) consumed 6.9 mL of titrant.

$$c(Ca^{2+}) = \frac{V_T * c_T}{V_S} * 1000 = \frac{6.9\,mL * 0.01\,mol/L}{50\,mL} * 1000 = 1.38 mmol/L$$

4) **Magnesium Concentration (c (Mg^{2+})):** The titration of 50 mL of raw water for Total Hardness (Ca + Mg) with 0.01 M EDTA (using Erichrome Black T) consumed 9.6 mL of titrant. First, the total concentration of (Ca + Mg) ions was calculated.

$$c(Ca^{2+} + Mg^{2+}) = \frac{V_T * c_T}{V_S} * 1000 = \frac{9.6\,mL * 0.01\,mol/L}{50\,mL} * 1000 = 1.92 mmol/L$$

And, Magnesium concentration was determined by subtraction:

$$c(Mg^{2+} = c(Total) - c(Ca^{2+}) = 1.92 - 1.38 = 0.54\, mmol/L$$

Data Summary of the Experiment

The data obtained from the raw water analysis and the data provided by the other groups for the four reactors are summarized in Tables as follows:

Parameter	Raw Water	HYDROLIT-Ca	Hydro-Carbonate	AKDOLIT-GRAN	MAGNO-DOL
pH	4.9	7.7	6.6	8.5	10.1
Temp (C)	21	20.8	21.3	20.5	20.6
Conductivity (uS/cm)	491	638	800	1072	906
ANC4.3 (mmol/L)	2.3	4	6.4	11.5	10.2
ANC8.2 (mmol/L)	-	-	-	-	2.2
BNC8.2 (mmol/L)	3.6	0.3	6.4	-	-
Ca (mmol/L)	1.38	1.64	3.42	0.62	0.28
Mg (mmol/L)	0.54	2.8	0.3	5.3	5.54

Hardness Classification of Treated Water					
Sample	Ca2+ (mmol/L)	Mg2+ (mmol/L)	Total Hardness (mmol/L)	Hardness Level	Classification
Raw Water	1.38	0.54	1.92	2	Medium Hard
HYDROLIT-Ca	1.64	2.8	4.44	4	Very Hard
Hydro-Carbonate	3.42	0.3	3.72	3	Hard
AKDOLIT-GRAN	0.62	5.3	5.92	4	Very Hard
MAGNO-DOL	0.28	5.54	5.82	4	Very Hard

The de-acidification process significantly increased the total hardness of the water (shifting from Medium Hard to Very Hard). This confirms that the filter materials dissolved into the water to neutralize the aggressive CO_2. Specifically, the Magnesium-containing filters (Akdolit and Magno-Dol) released high levels of Magnesium, whereas the Hydro-Carbonate filter released primarily Calcium and to a lesser extent."

Evaluation of the Calco-Carbonic Equilibrium

To evaluate whether the water samples were corrosive, balanced, or scaling, the theoretical equilibrium conditions were calculated using the physical constants provided in the lab script (K_{sp}, K_{a1}, K_{a2})

Ionic Strength and Activity

Calculation of Ionic Strength: $\mathbf{I} = \frac{\kappa_{25°C}}{62000} = 0.0079\ \mathrm{mol/L}$

Calculation of Activity Coefficient (Güntelberg): $\mathbf{\log(\gamma_1)} = -\mathbf{0.5}\left(\frac{\sqrt{\mathbf{I}}}{\mathbf{1+1.4\sqrt{I}}}\right) \Rightarrow \mathbf{\gamma_1} = 0.9129$

Tillman's Correction Factor: $\mathbf{f_T} = \frac{\mathbf{1}}{\gamma_1^6} = 1.727$

Tillman's Equilibrium (CO_2)

Tillman's Constant: $\mathbf{K_T} = \frac{\mathbf{K_{a2}}}{\mathbf{K_{sp}K_{a1}}} = 2.42 * 10\text{^}4$

Equilibrium CO_2 Calculation: $\mathbf{c(CO_2)_{eq}} = \left(\frac{\mathbf{K_T}}{\mathbf{f_T}}\right)\mathbf{c(HCO_3^-)^2 c(Ca^{2+})} = 0.044\ \mathrm{mmol/L}$

Langelier's Saturation Index (pH)

Langelier' Constant: $\mathbf{K_{La}} = \frac{\mathbf{K_{a2}}}{\mathbf{K_{sp}}} = 8.28 * 10^{-3}\ \mathrm{L/mol}$

Langelier's Correction Factor: $\boldsymbol{f_{\{La\}}} = \boldsymbol{\gamma_1^5} = 0.634$

Equilibrium pH:

$$\mathbf{pH_{eq}} = -\mathbf{log(K_{La})} - \mathbf{log(f_{La})} - \mathbf{log}\left(\mathbf{c}(\mathbf{HCO_3^-})\right) - \mathbf{log}\left(\mathbf{c}(\mathbf{Ca^{2+}})\right) = 7.78$$

Saturation Index: $\mathbf{S_I} = \mathbf{pH_{measured}} - \mathbf{pH_{eq}} = -2.88$

For MAGNO-DOL (pH = 10.1)

$\mathbf{c}(\boldsymbol{HCO})_{\mathbf{3\ eq}}^{-} = \boldsymbol{ANC_{4.3}} - \mathbf{2} * \mathbf{c}(\boldsymbol{CO_3^{2-}}) = 5.8\ \mathrm{mmol/L}$

Parameter	Raw Water	HYDROLIT-Ca	Hydro-Carbonate	AKDOLIT-GRAN	MAGNO-DOL
Ionic Strength (I) [mol/L]	0.01	0.01	0.01	0.02	0.015
Activity Coeff. (γ_1)	0.91	0.90	0.89	0.88	0.89
Tillman Factor (f_t)	1.73	1.85	1.97	2.15	2.04
Meas. CO_2 [mmol/L]	3.60	0.30	6.40	-	-
Equil. CO_2 [mmol/L]	0.04	0.09	0.27	0.08	0.02
Measured pH	**4.90**	**7.70**	**6.60**	**8.50**	**10.10**
Sum activity coefficient (fLa)	**0.63**	**0.60**	**0.57**	**0.53**	**0.55**
Equil. pH (pH_eq)	**7.78**	**7.49**	**6.99**	**7.51**	**8.13**
Saturation Index (SI)	-2.88	0.21	-0.39	0.99	1.97
State	Corrosive	Balanced	Corrosive	Scaling	Scaling

Constant	Symbol	Value	Unit	Description
Solubility Product	Ksp	3.93E-09	mol^2/L^2	Solubility limit of Calcite (CaCO3)
1st Dissociation	Ka1	3.43E-07	mol/L	Equilibrium of CO2 <-> HCO3
2nd Dissociation	Ka2	3.25E-11	mol/L	Equilibrium of HCO3 <-> CO3
Tillman's Constant	KT	2.42E+04	L^2/mol^2	Combined constant (Ka2 / Ksp * Ka1)
Langelier's Constant	Kla	8.28E-03	L/mol	Equals Ka2 / Ksp

SI Value	State of Water	Practical Meaning
SI<0 (Negatives)	Calcite-dissolving	Corrosive. The water has excess acidity and will dissolve the protective limestone layer in pipes.
SI = 0	Calcite-saturated	Balanced (Equilibrium). The water is stable. It neither dissolves pipes nor blocks them with scale. This is the goal of treatment.
SI >0 (Positive)	Calcite-precipitating	Scaling. The water is oversaturated. It will drop calcium carbonate out of solution, clogging pipes and heat exchangers.

Conclusion

The experiment demonstrated that chemical de-acidification of corrosive water depends on the type of filter material used. The initial raw water (minerals added) was highly corrosive (SI = -2.88) with an excess amount of free carbon dioxide (3.6 mmol/L).

- HYDROLIT-Ca ($\mathbf{CaCO_{3(s)} + 10\% MgCO_{3(s)}}$) proved to be the optimal material. It was successful in establishing calco-carbonic equilibrium (SI = 0.21) and produced water that complies with the chemical requirements of the German Drinking Water Ordinance. [2]
- Pure Calcium carbonate (Hydro-Carbonate) failed to neutralize the CO_2 in time due to its slow reaction time. With higher contact time, this material dissociates better.
- Dolomite Materials (MgO containing AKDOLIT-GRAN and MAGNO-DOL) were highly reactive but caused precipitating (scaling) conditions due to overshooting (SI > 0) and increased the pH values. PH measured as 10.1 exceeds the legal limit as per German Drinking Water

Ordinance. [2] This material functions better with faster flow rate or in conditions where only a part of the water (e.g. 50%) passes through it before mixing with the other portion.

Therefore, for the treatment of this particular water sample, HYDROLIT-Ca is recommended.

References

[1] Lab Script: Institute of Water Chemistry, TU Dresden. Lab Class MHSE 08 - Hydrochemistry, Experiment: Chemical De-acidification using calcium carbonate and semi-burnt dolomite.

[2] Trinkwasserverordnung: Verordnung über die Qualität von Wasser für den menschlichen Gebrauch (Trinkwasserverordnung) [Ordinance on the Quality of Water Intended for Human Consumption]. Germany. Parameters for chemical quality: Calcite dissolution capacity & Hydrogen ion concentrations